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Research Article

Algorithmic Classification of Five Characteristic Types of Paraphasias

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Purpose: This study was intended to evaluate a series of algorithms developed to perform automatic classification of paraphasic errors (formal, semantic, mixed, neologistic, and unrelated errors).

Method: We analyzed 7,111 paraphasias from the Moss Aphasia Psycholinguistics Project Database (Mirman et al., 2010) and evaluated the classification accuracy of 3 automated tools. First, we used frequency norms from the SUBTLEXus database (Brysbaert & New, 2009) to differentiate nonword errors and real-word productions. Then we implemented a phonological-similarity algorithm to identify phonologically related real-word errors. Last, we assessed the performance of a semantic-similarity criterion that was based on word2vec (Mikolov, Yih, & Zweig, 2013).

Results: Overall, the algorithmic classification replicated human scoring for the major categories of paraphasias studied with high accuracy. The tool that was based on the SUBTLEXus frequency norms was more than 97% accurate in making lexicality judgments. The phonological-similarity criterion was approximately 91% accurate, and the overall classification accuracy of the semantic classifier ranged from 86% to 90%.

Conclusion: Overall, the results highlight the potential of tools from the field of natural language processing for the development of highly reliable, cost-effective diagnostic tools suitable for collecting high-quality measurement data for research and clinical purposes.

In people with aphasia, the ability to produce words is typically impaired, resulting in anomia (Goodglass & Wingfield, 1997). In single-word retrieval, as described in models of word production (e.g., Dell, 1986; Levelt, Roelofs, & Meyer, 1999), this deficit is believed to be indicative of disruption in activating the relevant semantic features of the target concept and/or retrieving a fully phonologically specified representation (e.g., Dell, 1986). Aphasic naming errors are generally considered to result from reduced or insufficiently persistent activation of target representations relative to competing nontarget representations and/or noise in the system (Schwartz, Dell, Martin, Gahl, & Sobel, 2006). Depending on the severity of the lesion and its locus within the model, different error patterns may result. Reduced activation of lexical-semantic representations may result in a larger proportion of semantic errors (e.g., *dog* for the target *cat*). Form-related words may also become activated because of spreading activation and feedback (e.g., *mat* for *cat*). During the second step, activation of inappropriate phoneme representations may result in more frequent phonological errors (neologisms; e.g., *tat* for the target *cat*) or real-word phonemic errors (e.g., *dog* for the target *log*; Foygel & Dell, 2000). In addition, mixed semantic and phonological errors may occur (e.g., *rat* for *cat*).

One commonly used method for assessing anomia severity in people with aphasia is the use of confrontation naming tests. One such test is the Philadelphia Naming Test (PNT; Roach, Schwartz, Martin, Grewal, & Brecher, 1996). Fergadiotis, Kellough, and Hula (2015) and Hula, Kellough, and Fergadiotis (2015) recently used item response theory (Lord & Novick, 1968) to calibrate the PNT using a one-parameter logistic model. Using a real-data simulation, they then tested a computer adaptive version of the PNT (Dorans et al., 2000). In the current computer adaptive implementation of the test, the examining clinician manually enters a dichotomous correct–incorrect judgment for the naming response of the test taker after each item presentation, and given that input, the algorithm re-estimates the test taker's ability. Then, given the item-difficulty parameters (Fergadiotis et al., 2015), the algorithm selects the optimal item to administer next by ignoring items that are judged to be too easy or too hard and thus uninformative for a given test taker. The technique quickly

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converges into a sequence of items bracketing the test taker's ability, thus efficiently shortening the test while maximizing its precision. Using the computer adaptive algorithm, Hula et al. (2015) reported that a 30-item adaptive PNT form correlated highly (Pearson product–moment correlation = .95) with the full 175-item PNT.

However, even though scores derived from tests with a unidimensional scoring system are useful in scaling and ordering subjects along a severity continuum, these proficiency scores contain limited diagnostic information necessary for the identification of subjects' specific strengths and weaknesses. Error profiles have been used extensively in research investigations and are crucial for increasing our knowledge of neurotypical and impaired word production and for developing effective clinical practices. Dell's theoretical framework has been formalized as a computational model, and anomic error profiles have been used extensively to test competing hypotheses regarding the architecture and information processing of the cognitive machinery underlying word production (Dell, 1986; Dell & O'Seaghdha, 1992; Dell, Schwartz, Martin, Saffran, & Gagnon, 1997; Nozari & Dell, 2013; Schwartz et al., 2006). Further, anomic error profiles are often used for the development and evaluation of new diagnostic and treatment approaches. For example, error profiles have been used to classify people with aphasia into subgroups depending on the nature of their underlying anomic deficits prior to investigating the generalization of anomia treatment as a function of subgroup characteristics (Best et al., 2013); Kendall, Pompon, Brookshire, Minkina, and Bislick (2013) evaluated the efficacy of the phonomotor treatment by investigating whether the intervention was associated with systematic shifts in the error distributions of anomic error profiles that would suggest changes in linguistic processing; Edmonds and Kiran (2006) focused on the evolution of errors to study the effects of a semantic anomia treatment on cross-linguistic generalization; and Fridriksson, Richardson, Fillmore, and Cai (2012) associated patterns of cortical reorganization following anomia treatment that were related to specific types of errors.

These profiles could also be clinically useful for developing individualized intervention plans (Abel, Willmes, & Huber, 2007), but classifying errors from confrontation naming tests is time consuming and often it is not performed in fast-paced clinical settings. Therefore, there is a need to develop efficient and psychometrically robust tools to quantify impairment along more than a single dimension. One of the challenges in achieving this goal is the need to administer a large number of items to derive stable estimates that would allow for meaningful interpretations regarding a subject's underlying cognitive deficits (Walker & Schwartz, 2012). The reason is that static tests have to include a sufficiently large number of items to be informative for the majority of people with varying severity levels. To address this issue, it is possible to utilize multidimensional item response theory (de la Torre & Patz, 2005; Embretson, 1998) to build a cognitive diagnostic model for measuring concurrently multiple constructs (e.g., lexical-semantic and phonological processing) using a single set of items. However, a significant barrier to creating an efficient computer adaptive tool would be scoring responses in an online manner, so as to provide the algorithm with input for making a selection after each word production. This would in turn allow the algorithm to dynamically choose which items to administer on the basis of what would be most informative for each subject. This would be critical for developing a test that would efficiently converge to stable error distributions without requiring the administration of all the items in a test. The successful implementation of the computer adaptive PNT with unidimensional scoring (Hula et al., 2015) critically depends on the ability of the examining clinician to make online judgments only about the accuracy of the responses. However, the successful implementation of a computer adaptive test with multidimensional scoring would work only if errors were classified in distinct categories immediately after the production of each paraphasia.

Such a task would be very challenging. Current approaches for developing a subject's profile depend on human annotators to compare productions and target words, determine the lexical status of a production, make semantic associations, and use phonological-similarity criteria. Even when errors are classified by experienced professionals offline and in the absence of time constraints, reliability estimates indicate that this is a difficult task. For example, Minkina et al. (2016) reported an interrater reliability of 86% on paraphasic-error classification, which corresponded to a κ of .73 (taking into account agreement due to chance). Intrarater reliability was 86% for point-to-point agreement and yielded a κ of .76. Similar results have been reported in other recent investigations (Kristensson, Behrns, & Saldert, 2015; Leonard et al., 2015). Even when responses are from neurotypical participants, reliability estimates reflect difficulties in making classification judgments, especially with respect to the semantic relatedness of production and targets (Nicholas, Brookshire, Maclennan, Schumacher, & Porrazzo, 1989). These estimates can be considered upper bounds when attempting to extrapolate the reliability in clinical settings because of the presence of time constraints and the lack of an opportunity to consult with colleagues and reach a consensus for problematic responses. The development of automatic classification algorithms to assist in alleviating these problems is the main motivation for this study.

In addition to developing algorithms for computer adaptive confrontation naming tests, technologies that would allow a machine to "understand" the nature of anomic errors could be used for many other applications. For example, paraphasia classification algorithms can be used to annotate errors in large databases including AphasiaBank (MacWhinney, Fromm, Forbes, & Holland, 2011) and the Moss Aphasia Psycholinguistics Project Database (MAPPD; Mirman et al., 2010). In addition to the efficiency and reliability introduced by an automated process, algorithmic classification has the advantage of requiring developers to explicitly state the rationale underlying the classification criteria and to hardwire it into the code. This

includes both the well-defined aspects of the algorithm and the algorithm's simplifying assumptions. This in turn maximizes the transparency of the coding process and allows for maximum reproducibility of results. It can also allow for rapid recoding of extant data sets as our understanding of anomia advances and classification criteria are updated.

The field of natural language processing (NLP) is a branch of computer science that focuses on human speech and language. It covers a very wide range of linguistic tasks, from the relatively low level (developing computational models of verb inflection, identifying a word's part-of-speech status, etc.) to highly complex and sophisticated linguistic analyses such as anaphora resolution, temporal-relation extraction, and question answering (Hirschberg & Manning, 2015). NLP finds applications in a wide variety of areas, including search engines such as Google and speech-recognition systems such as Apple's Siri or Amazon's Alexa. It also has numerous biomedical applications (for a recent review, see Névéol & Zweigenbaum, 2015); of particular relevance to the present article is work applying NLP to analysis of the language produced by individuals with (or at risk of) neuropsychological conditions. Much of this work has had as its goal the early detection or characterization of disorders such as Alzheimer's disease (Roark, Mitchell, Hosom, Hollingshead, & Kaye, 2011) and primary progressive aphasia (Fraser et al., 2014). Other research in the field, notably, has focused on using NLP techniques and tools to automate the administration and/or scoring of neuropsychological assessments involving language (Prud'hommeaux & Roark, 2011). To our knowledge, ours is the first work to specifically apply NLP techniques to the problem of analyzing individual paraphasic speech errors themselves, rather than speaker-level categorization. There has, however, been much work done within the field of NLP on problems such as the detection of grammatical errors, particularly in the context of language learning (Leacock, Chodorow, Gamon, & Tetreault, 2010). There has also been work in the area of computerized pronunciation analysis, again aimed at language learners (Moustroufas & Digalakis, 2007) and children (Dudy, Asgari, & Kain, 2015).

The specific techniques we used in this study were chosen to map to several of the discrete subtasks that a human must perform when scoring a naming test: determining a production's lexicality and assessing the production's phonemic and semantic similarity to a target word. Of these, the most computationally challenging is that of assessing semantic similarity. We used a statistical approach called *word2vec* (Mikolov, Yih, & Zweig, 2013; see the Method section for additional details) that, though relatively new, has proven useful for tasks such as analogy and sentence completion.

Purpose of the Study

The present study focused on the development and evaluation of a series of algorithms developed to perform automatic classification of paraphasic errors according to an already-existing typology. The specific research questions of this study were the following: (a) How accurately can we determine the lexicality of paraphasic errors to differentiate neologisms from real-word errors using an automated approach? (b) How accurately can we differentiate formal and mixed errors from semantic errors, unrelated errors, and perseverations on the basis of an automated phonological-similarity criterion? (c) How accurately can we differentiate semantically related errors (semantic, mixed) from semantically unrelated errors (formal, unrelated, perseverations) on the basis of an automated semantic-similarity criterion?

Method

Participants and Data Set Preparation

The data analyzed for this study consisted of 7,111 errors retrieved from MAPPD (Mirman et al., 2010).[1] Data consisted of the error responses of 251 participants on the PNT. All participants were community-dwelling individuals with aphasia consequent to left-hemisphere stroke, were right-handed, were native English speakers, and had no comorbid neurologic illness or history of psychiatric illness. Demographic data and descriptive statistics are provided in Table 1. With respect to the PNT items, the following psycholinguistic variables can be found in Fergadiotis et al. (2015, Supplemental Text B): (a) length in number of phonemes, (b) rated age of acquisition in years (Kuperman, Stadthagen-Gonzalez, & Brysbaert, 2012), and (c) SUBTLEXus frequency norms (Brysbaert & New, 2009). Additional information on the psycholinguistic characteristics of the PNT items can be found in Roach et al. (1996). In addition, brief definitions and distributions for the different categories of errors can be seen in Table 2.

It is noteworthy that additional information regarding the human annotators and their reliability was not readily available because MAPPD data have originated from multiple different published data sets in addition to some unpublished data. However, MAPPD data have been used extensively over the past two decades, yielding consistent and reliable results (for more information, see Mirman et al., 2010).

Human annotators from MAPDD provided a phonemic transcription of each error and coded the error according to the PNT guidelines (Roach et al., 1996) and the intended target. For this analysis we focused on five major categories of paraphasias: formal, nonword, semantic, mixed, and unrelated paraphasias. These categories were selected because they constitute the vast majority of errors

[1]A note on sample size: The full MAPPD data set consists of 7,111 productions. For the various analyses described in this article, we produced subsets of the entire MAPPD data depending on the specific needs of each analysis. First, though, we excluded 110 productions in which the patient produced multiple words (e.g., "bus doctor"). Therefore, the true starting point for all analyses was actually 7,001 paraphasias. A detailed description of the data-preparation process can be found in Supplemental Material S1.

Table 1. Demographic and clinical characteristics of the participant sample.

Characteristic	Value
Ethnicity (%)	
African American	34
Asian	> 1
Hispanic	1
White	44
Missing	20
Education (years)	
M (*SD*)	13.6 (2.8)
Minimum	7
Maximum	21
Missing (%)	20
Age (years)	
M (*SD*)	58.8 (13.2)
Minimum	22
Maximum	86
Missing (%)	20
Months post onset	
M (*SD*)	32.9 (51.0)
Minimum	1
Maximum	381
Missing (%)	20
Western Aphasia Battery–Revised Aphasia Quotient (Kertesz, 2007)	
M (*SD*)	73.4 (16.6)
Minimum	27.2
Maximum	97.8
Missing (%)	51
Philadelphia Naming Test (% correct)	
M (*SD*)	61 (28)
Minimum	1
Maximum	98

in the database and they can be used as input for Dell's connectionist model to derive connectionist weights that reflect the impairment of phonological and semantic subprocesses (Saffran, Dell, Schwartz, & Gazzaniga, 2000).

We converted the phonetic transcriptions from MAPDD (which use the International Phonetic Association guidelines for segments and thus lacks stress information) to a different transcription system, known as ARPAbet. ARPAbet is widely used in speech recognition and synthesis, and we were able to make use of a large, digitized, freely available pronunciation dictionary (the Carnegie Mellon University Pronouncing Dictionary [CMUdict]) which makes use of the ARPAbet convention. Once production transcriptions have been converted to ARPAbet, it is possible to look them up in CMUdict to determine their likely orthographic form—assuming the productions are not neologisms—which can then be used to look up word frequency.

Algorithmic Lexicality Analysis

We first attempted to discriminate between nonword and real-word errors. The former included neologisms that were phonologically related to the target and abstruse neologisms. The latter included semantic errors, formal errors, mixed errors, unrelated errors, and perseverations. According to the published guidelines of the PNT, to classify a production as a real word, human annotators of the MAPPD database used *Merriam-Webster's Collegiate Dictionary*. For this analysis, to determine the lexicality of each production, we first queried the SUBTLEXus database (Brysbaert & New, 2009), which was compiled from the subtitles for 8,388 U.S. films and television episodes and comprises 51 million words in all. Productions not found in SUBTLEXus were all tagged as nonword errors. Brysbaert and New (2009) compared the SUBTLEXus frequency norms with frequency norms from the Brown corpus (Kučera & Francis, 1967), CELEX2 (Baayen, Piepenbrock, & Gulikers, 1996), and HAL (Burgess & Livesay, 1998). They reported that, compared with these older norms, the SUBTLEXus norms are more closely correlated with a number of psycholinguistic behavioral measures—including naming latencies—in typical populations. We further investigated the utility of word frequency to identify nonword errors that were phonetically identical to rare English words (e.g., *bray* [bɹeɪ]). Henceforth, consistent with Butterworth (1979), we refer to these productions as *jargon homophones*. Jargon homophones can, at times, be systematically misclassified as true-word productions because their phonetic transcription matches entries in databases or dictionaries. The human annotators that classified the errors in MAPPD did not differentiate

Table 2. Error distribution and definitions.

Error type	*n*	Definition
Formal	1,459	Real-word errors that satisfied the phonological-similarity criterion.
Neologisms	2,002	Nonword responses that *are* phonologically related to the target.
Abstruse neologisms	658	Nonword responses that *are not* phonologically related to the target.
Semantic	1,130	Real-word errors that are semantically related to the target.
Mixed	652	Real-word errors that are both semantically and phonologically related to the target.
Perseverations	600	Responses that were produced by the participant on a previous trial within the same session.
Other	610	Real-word errors that are not semantically or phonologically related to the target; includes visual errors.
Total *N*	7,111	

Note. Phonological relatedness was based on the phonological-similarity criterion.

this subset of errors explicitly. However, psychologically, their classification as true-word errors may misrepresent the source of the errors; specifically, it incorrectly suggests a breakdown in lemma access rather than a breakdown in phonological encoding (Dell, 1986).

We hypothesized that jargon homophones would have lower frequency than real-word errors such as semantic and formal errors. To test this hypothesis we used the frequency norms from the SUBTLEXus database as decision thresholds for a binary classifier. To be more specific, the class prediction for each production was made on the basis of frequency X. Given a threshold parameter T, the production was classified as a real-word error if $X > T$ (i.e., if the word's frequency was greater than T) and a nonword error otherwise.

To analyze the utility of word frequency as a classifier, we used receiver operating characteristic (ROC) curve analysis (Macmillan & Creelman, 1991). ROC curves are commonly used to explore the performance characteristics of signal detection and classification algorithms in fields ranging from medical diagnostic testing to radar engineering. An ROC curve plots a classifier's true-positive rate against its false-positive rate over the classifier's range of possible operating points (in this case, frequency-value thresholds). The ROC curve, then, visualizes the effect of changing our classifier's frequency cutoff value on performance.

To evaluate the discriminant utility of this classifier, we calculated the area under the ROC curve (AUC), a common metric for evaluating binary classification (Hanley & McNeil, 1982). This metric, notably, can be used to evaluate any binary classifier and is independent of the frequency threshold. In this application, it measured the probability that a randomly chosen real-word error would be of higher frequency in the SUBTLEXus corpus than a nonword error. We also evaluated the performance of the classifier at its optimal operating point using sensitivity (i.e., true-positive rate) and specificity (i.e., true-negative rate). For this analysis, a true positive was a production that (a) was coded by the MAPPD human annotators as a real-word error (i.e., formal, semantic, mixed, unrelated) and (b) was also decided by our automated classifier to be a real word. To identify the classifier's optimal operating point (i.e., to choose the threshold to use for the sensitivity and specificity analysis), we identified the frequency value that maximized the harmonic mean of the two metrics (F1 sensitivity and F1 specificity, respectively).

Phonological-Similarity Analysis

Next, we removed the nonword productions from the analysis[2] and tested an automated method designed to identify real-word errors that were phonologically related to the target. To achieve this, we built an algorithm which attempts to mimic the phonological-similarity criterion from the PNT scoring guidelines. According to these guidelines, an error was classified as phonologically similar to the target if it shared one of the following characteristics with the target: (a) the stressed vowel or initial or final phones, (b) two or more phones (excluding unstressed vowels) at any position, or (c) one or more phones at a corresponding syllable and word position, aligning words from left to right. Our algorithm takes as its input phonemic transcriptions of the target and production. It first uses a basic set of linguistic rules to syllabify the target and the production (Gorman, 2013). It then compares the two, syllable by syllable and phone by phone, for each of these three rules and indicates whether the transcriptions match on any of the three. For instance, if the target is *carrot* [kæɹət] and the production is *rabbit* [ræbɪt], the algorithm will indicate that they match both on stressed vowel [æ] and final phone [t], and therefore are to be coded as phonologically similar. Unlike frequency, this was an inherently binary classification problem—there is no need to select an operating point—and to evaluate the algorithm we calculated classification accuracy, sensitivity, and specificity.

Semantic-Similarity Analysis

For our final step, we attempted to discriminate between semantically related and semantically unrelated errors. This analysis was performed twice on different subsets of the data and with different goals. First, the semantic classifier was applied to the subset of errors that were phonologically related to the target. This subset included mixed errors (e.g., *rat* for *cat*) and formal errors (e.g., *pat* for *cat*). By definition, only mixed errors share a semantic similarity with the target. The semantic classifier was subsequently applied to a subset of errors, including ones that were phonologically unrelated to the target. This subset included semantic errors (e.g., *dog* for *cat*), unrelated errors (e.g., *chair* for *cat*), and perseverations. Only the first category of errors shares a semantic similarity with the targets. We decided not to perform a single analysis by aggregating across semantically related and unrelated errors (i.e., mixed and semantic vs. formal and unrelated), to allow for a more detailed analysis and investigate whether the accuracy of the semantic classifier interacted with phonological similarity.

We modeled semantic similarity using word2vec (Mikolov et al., 2013), a relatively new machine-learning technique (that is based on recurrent neural networks) that analyzes large amounts of textual data and, by observing patterns of word use and co-occurrence, develops a model of the semantic relationships between pairs of words. Given appropriate data, it can learn, for example, that *car* and *truck* are more closely related than *car* and *banana*, and it can yield continuous values that reflect the magnitude of similarity (expressed as the cosine similarity between the vector-space embeddings of the production and the target words). We trained a word2vec model using the Gigaword corpus (Parker et al., 2011) of newswire text as well as a locally developed corpus of transcripts of a

[2]See Table 3 for a detailed description of precisely which errors were removed for this part of the analysis.

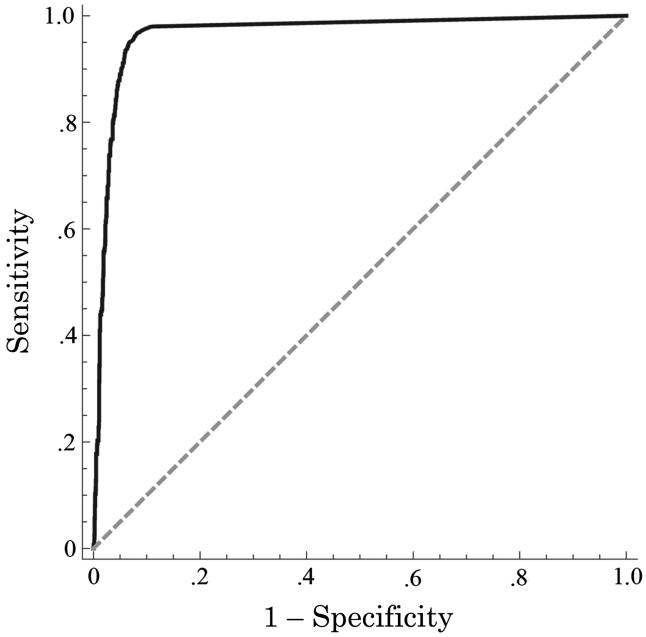
long-running U.S. public radio program. The Gigaword corpus was chosen because it is one of the largest publicly available databases of English text and is, when compared with "web text" sources, much more carefully edited, because the text is produced by professional journalists. We included our public-radio corpus to attempt to expose our model to examples of more natural and conversational language. We once again used ROC curves and AUC to measure discrimination; here, AUC represents the probability that a randomly chosen error classified by human annotators as mixed or semantic will be more semantically similar to its target than will a randomly chosen formal or unrelated error. Sensitivity and specificity statistics were also estimated.

Results

Lexicality Analysis

After the exclusion of multiword productions and words that could not be trivially converted to ARPAbet, 7,001 errors were analyzed to determine the performance of the automated lexicality classifier. With respect to the discrimination of nonword and real-word errors, approximately 89% were classified accurately on the basis of the SUBTLEXus frequency norms without taking into account the frequency of the error productions. However, as hypothesized, we found frequency of the production to be a highly discriminative cue. An ROC curve showing the trade-off between sensitivity and specificity can be seen in Figure 1. For this analysis, sensitivity reflects the probability of correctly classifying real-word errors as the frequency threshold is varied (*y*-axis). The *x*-axis reflects the false-positive rate (i.e., 1 − specificity), where *specificity* refers to the probability of correct classification of nonword errors. A high frequency threshold would lead to correct classification for the vast majority of nonword errors (i.e., high specificity) at the expense of low accuracy for the classification of real-word errors (i.e., low sensitivity). In contrast, a low frequency threshold would ensure high accuracy for the classification of real-word errors (i.e., high sensitivity) to the detriment of the classification accuracy for nonwords (i.e., low specificity). The closer the curve follows the left-hand border and then the top border of the ROC space, the more accurate the classifier; on the other hand, the closer the ROC curve approximates the 45° diagonal of the ROC space, the less accurate the classifier. A visual inspection of the figure and the AUC estimate (AUC = .975, 95% confidence interval [CI] [.963, .971], $SE = 0.002$) are consistent with the highly discriminant utility of this classifier. In conceptual terms, the AUC corresponds to the probability that a randomly drawn real-word error has a higher frequency compared with a jargon homophone. To identify the optimal threshold, we estimated the frequency value that "balanced" sensitivity and specificity (i.e., F1 frequency threshold). The optimal threshold was determined to be 11—that is, words that occurred more than 11 times in the SUBTLEXus corpus (approximately once per five million words) "counted" as a word, and otherwise were labeled as a nonlexical production. F1 sensitivity was .981, and F1 specificity was .919.

Figure 1. Receiver operating characteristic (ROC) curve for real words versus nonwords, using a frequency criterion. Sensitivity reflects the probability of correctly classifying real-word errors. The *x*-axis reflects the false-positive rate, where specificity refers to the probability of correct classification of nonword errors. The ROC curve reflects the trade-off between sensitivity and specificity: As a lower frequency threshold is chosen, more real-word errors are classified correctly at the expense of correct classification of nonword errors.



Phonological-Similarity Analysis

Our mechanistic approach for identifying errors that were phonologically related to the target was also highly effective. We used 4,365 errors for this analysis, after nonwords and morphologically related errors were excluded. The results are presented in Table 3. This table allows visualization of the performance of the algorithm; each column represents the instances in a predicted error category (i.e., phonologically similar vs. not phonologically similar), and each row represents the instances in an actual class on the basis of the a priori human-annotator classification. The overall accuracy of this classifier was .911. The sensitivity of the classifier that reflects the probability of correctly identifying phonologically related errors was .914. The specificity of the classifier that captures the probability of correctly identifying phonologically unrelated errors was .907.

Semantic-Similarity Analysis

Our results suggested that semantic similarity as quantified by word2vec was highly discriminative. Figure 2 presents two ROC curves. The solid ROC curve represents the trade-off between sensitivity and specificity as a

Table 3. Confusion matrix for phonological-similarity criterion.

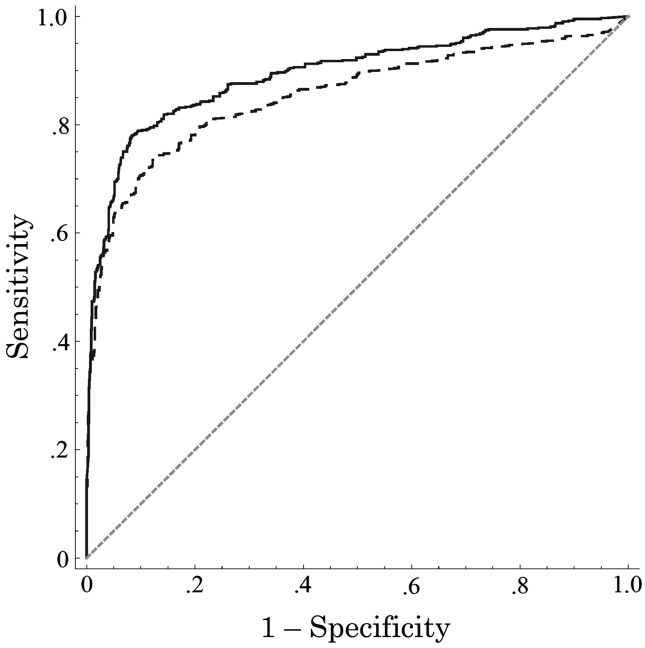
Coder judgment	Predicted: Phonologically similar	Predicted: Not phonologically similar
Coder: Phonologically similar	1,872	177
Coder: Not phonologically similar	181	1,981

Note. The values on the diagonal reflect true positives (1,872) and true negatives (1,981). The values on the off-diagonal reflect false positives (181) and false negatives (177). Also note that this analysis was conducted on a total of 4,211 errors. Our overall data set consisted of 7,111 productions, but for this analysis we excluded multiword productions, neologistic productions, and productions that were morphologically similar to their target word.

function of all the possible semantic-similarity thresholds on the basis of which the algorithm could differentiate 2,049 mixed and formal paraphasias. AUC was .895 (95% CI [.877, .912], SE = 0.009), suggesting that the semantic algorithmic classifier was very accurate in differentiating errors on the basis of semantic similarity. To identify the optimal threshold, we estimated the semantic-similarity value that "balanced" sensitivity and specificity (i.e., the F1 semantic-similarity threshold). At the F1 semantic-similarity threshold, the probability of correctly identifying mixed errors (sensitivity) was .789 and the probability of correctly identifying formal errors was .906.

The ability of the semantic classifier to differentiate semantic errors from unrelated errors (N = 2,162) is captured by the dashed ROC curve in Figure 2. Again, the semantic classifier performed well (AUC = .861, 95% CI [.838, .870], SE = 0.009). F1 sensitivity (i.e., the probability of correct classification of semantic errors using the F1 threshold that "balances" sensitivity and specificity) was .767. F1 specificity that represents the probability of correctly classifying an unrelated error was .919. For this analysis, the threshold for determining whether an error was semantically related to the target or not was estimated to be .56. Table 4 shows several examples of false-positive and false-negative semantic-similarity judgments.

Figure 2. Receiver operating characteristic (ROC) curve using the word2vec semantic-similarity scores. The solid curve corresponds to the ROC curve constructed using semantic similarity to differentiate mixed and formal paraphasias. The dashed curve corresponds to the ROC curve for differentiating semantic and unrelated errors.



Discussion

The main motivation of this study was the development of a series of algorithms for automatic aphasic error classification during confrontation naming tests. Overall, the scoring algorithms described in this article replicated human scoring for the major categories of paraphasias with high accuracy.

Lexicality Analysis

The use of the SUBTLEXus database (Brysbaert & New, 2009) was effective for differentiating real-word errors from nonword errors. The disagreements that did occur between our mechanistic approach and the results reported in the MAPPD can be attributed to two factors. First, the MAPPD human annotators used *Merriam-Webster's Collegiate Dictionary* rather than corpus frequencies. Second, we used a frequency criterion determined by ROC-curve analysis to identify and correctly classify jargon homophones. As a result, a number of productions with very low frequency that were classified as real words by the human annotators were judged by our algorithm to be neologistic productions that happened to resemble real words. We argue that our approach is more realistic because it does not assume that each subject possesses a lexicon as expanded as *Merriam-Webster's Collegiate Dictionary* (D'Anna, Zechmeister, & Hall, 1991; Goulden, Nation, & Read, 1990; Nation & Waring, 1997). The direct clinical implication of these findings is that professionals should avoid the use of the simple heuristic often used for the classification of neologistic errors by referring to dictionaries; instead, considering the frequency of occurrence of the production should be routinely factored in when determining whether a production is a real-word error or a neologism. This would, of course, make the scoring process substantially more complex, but it also represents an excellent example

Table 4. Examples of semantic false positives and false negatives.

Coder judgment	Predicted: Semantically similar	Predicted: Not semantically similar
Coder: Semantically similar	True positives (n = 1,264)	False negatives (n = 411) T: *seal*; P: *tiger* T: *ghost*; P: *spook* T: *van*; P: *bike* T: *spider*; P: *arachnophobia* T: *table*; P: *stool* T: *saw*; P: *sword*
Coder: Not semantically similar	False positives (n = 276) T: *pickle*; P: *dice* T: *pipe*; P: *pump* T: *nose*; P: *nail* T: *chimney*; P: *brick* T: *bread*; P: *brie* T: *spoon*; P: *soap*	True negatives (n = 2,260)

T = target; P = production.

of how automated or computer-assisted scoring could enable new and more robust approaches to aphasia assessment.

That said, even though we feel that the use of the frequency criterion is an improvement over the criterion of the PNT scoring guidelines, we also recognize a limitation in our approach. The selection of the cut-point score on the ROC curve was determined in such a way as to maximize agreement with the human annotators even though the latter ignore jargon homophones. In the absence of a gold standard for explicitly identifying jargon homophones, our selection of the frequency criterion does not fully address the issue of differentiating true real-word errors from productions that are not part of a subject's mental lexicon but happen to assume the form of a real word by chance. Even though it is also perhaps infeasible at this point to construct a gold standard for the classification of jargon homophones, our approach reduces the probability of misclassification by identifying productions that are highly unlikely to be true-word errors.

It should also be noted that frequency of occurrence is commonly taken into consideration in selecting naming items for confrontation naming tests. However, in these cases it is factored in for a different purpose; specifically, the purpose of controlling for frequency in batteries such as the Comprehensive Aphasia Test (Swinburn, Porter, & Howard, 2004) is to keep item difficulty constant or increase it incrementally in a structured way. Through manipulating these variables, specific hypotheses can be tested regarding the cognitive deficits that underlie the symptoms of people with aphasia (e.g., Martin, 2017). Our use of frequency was for the different purpose of classifying errors as phonemic errors versus neologistic productions.

Phonological-Similarity Analysis

With respect to the phonological-similarity criterion, our algorithmic implementation proved highly successful. In point of fact, the estimates we reported for the algorithmic phonological analysis might be higher if we address two shortcomings in the future. First, we had to convert the phonetic transcriptions available in MAPPD (International Phonetic Alphabet) to a different phonetic transcription system commonly used in speech-language technologies and resources (ARPAbet). As a result, the human annotators and the machine used slightly different input when making their judgments. In addition, all of the analyses we report in this article assume that the human annotators are perfect in classifying errors. However, this is an unrealistic assumption, because human annotators may misclassify productions because of fatigue and lapses in attention or misapplication of the phonological-similarity rules. This may be a considerable source of both false positives and false negatives in our analyses (181 and 177 errors, respectively; see Table 3).

We further identified two systematic failures in our computer program that caused it to produce systematically incorrect results leading to an increased rate of false positives. First, consistent with the scoring guidelines of the PNT, human annotators ignored inflection for number. For example, the production *shoes* in response to the target *slippers* was not coded as phonologically similar by human annotators, even though it satisfied the first rule of the phonological-similarity criterion (i.e., the target and production shared a final phoneme). However, we do not currently have a reliable way to detect the presence of the regular plural suffix shared by the target and production, so we incorrectly labeled this example. The second challenge was related to the difficulty of applying the PNT phonological-similarity criteria to the MAPPD phonetic transcriptions, because the latter lack stress information on which the former depends. Although we are usually able to restore word stress by matching the MAPPD transcriptions against the ARPAbet transcriptions in CMUdict—which does mark word stress—we recommend that future work provide stress-marked transcriptions and use transcription guidelines that are compatible with digital resources such as CMUdict. Even though the current

analysis has an overall accuracy of approximately 91%, which is considered high, future analysis using digital transcription standards such as ARPAbet will allow us to fully explore the potential of this automated tool. A final point related to the phonological-similarity criterion pertains to its potential for identifying abstruse neologisms. We did not test this directly, but given the classifier's performance at correctly classifying phonologically unrelated errors (approximately 90%), we would predict that the classifier would be highly effective at this task as well.

Semantic-Similarity Analysis

The implementation of the semantic-similarity criterion also yielded very promising results. The machine-learning technique we used, word2vec, infers semantic relationships between words by analyzing word context and co-occurrence from a large corpus of text, and has lately been used to perform numerous tasks that require sophisticated models of semantic similarity (e.g., Amunategui, Markwell, & Rozenfeld, 2015; Le & Mikolov, 2014; Wolf, Hanani, Bar, & Dershowitz, 2014). We trained a word2vec model on a corpus of text containing both newswire text (i.e., text that originated in a written form) and transcribed text from a long-running radio program, meaning that our model contained information about word usage in a spoken, conversational context. This classifier was quite successful at performing the challenging task of making approximately the same decisions about semantic relationships between words as human annotators.

We hypothesize that disagreements between our semantic classifier and the human annotators who performed the MAPPD coding are the result of least three factors. First, there is intrarater variability in making semantic judgments. In addition, when multiple annotators are involved in a project, interrater variability is infused in the coding. For example, given a human annotator's past experiences, *guitar* (target) and *cowboy* (production) may or may not be coded as being semantically related. Both intra- and interrater sources of unreliability introduce noise in the coding that consequently blurs the relationship between our criterion (human coding) and our classifier. However, it is also noteworthy that the ability of human annotators to calibrate their scoring to individual differences can be a great advantage in situations when the semantic similarity of a target and a production changes as a function of the experiences of the subject. Human testers can flexibly adapt to such differences, thus personalizing their scoring appropriately. The third factor is related to the word2vec algorithm itself. Although it does represent one of the most effective known computational models of word similarity, it is by no means perfect, and it may be the case that our approach failed to capture certain semantic relations. Future studies will focus on whether there are undesirable patterns in how the machine classifies semantically related errors and will explore different approaches to computational modeling of semantic similarity.

Conclusions, Limitations, and Future Directions

Overall, this preliminary set of results highlights the potential of the tools developed in the field of natural language processing for the development of highly reliable, cost-effective diagnostic tools suitable for collecting high-quality measurement data for research and clinical purposes. It is noteworthy that the mechanistic tools presented in this article are not restricted to the PNT. For example, the phonological-similarity criterion we used mirrored the "liberal" guidelines of the PNT (Schwartz et al., 2006), and designing algorithms to mimic the phonological scoring criteria of other naming tests will be even more straightforward. Also, the semantic classifier can be used across tests as long as the scoring guidelines do not require the user to specify the nature of the semantic relationship (e.g., subordinate, superordinate).

We see two primary advantages to using an automated approach for semantic-similarity judgment. First, using an automated system to perform this task dramatically reduces the uncertainty associated with different human annotators scoring responses and allows research data to be generated under uniform procedures. Second, the recoding of large corpora of error productions can be simplified substantially (e.g., when updated versions of tests with different guidelines emerge). Another useful property of our automated scoring approach is that our classifiers can be designed to output the actual quantitative values underpinning their decisions (word frequency, word2vec semantic-similarity scores specifying which phonological-similarity rule was triggered, etc.) in addition to simply reporting paraphasia error codes. This would represent a valuable stream of secondary data about error productions that could itself be used as input to predictive models or decision-support systems. Our results suggest that it is possible to successfully implement automatic error classification for the development of computer adaptive tests in the future. However, there are still several issues that need to be addressed.

First, even though we were able to build algorithms to capture the major categories of errors represented in MAPPD, there were several small categories of clinically relevant errors that were ignored for the purposes of this article. To be specific, we did not develop a procedure to identify perseverative errors, and we did not attempt to differentiate between abstruse neologisms and neologisms that were phonologically related to the target. The former can be addressed in the future by designing a heuristic rule according to which a production will be flagged as a perseveration if a response to a new item matches a previous production during the same administration. The latter can be addressed by applying the phonological-similarity criterion to nonword errors. In addition, our methods did not focus on errors that were visually related to the target, nor did we test the performance of our semantic classifier at accurately classifying proper-noun responses that had a semantic relationship to the target and should be coded as semantic. A potential inexpensive solution to these problems

could be the use of crowdsourcing to build concordances of visual and semantic information from which a specialized algorithm could draw information. Because our semantic model is trained from text, it conflates a number of disparate forms of similarity, including pure associations (e.g., *wedding* and *cake*), but an anonymous reviewer pointed out that semantic errors in confrontational naming tasks are most likely to show category similarity (*dog* and *horse*). In future work, we hope to take advantage of NLP techniques that do distinguish between these sorts of similarities to further improve our classification of semantic errors. A final additional complicating factor is that the motoric nature of the errors was not taken into account at this point.

It is important to emphasize that even though the classification algorithms presented in this article appear robust, they do require phonetic transcriptions as input. In a sense, our system at this point can "read" pretty well and make decisions, but it cannot "hear" the productions. At this point, this is the main limiting factor before a fully automated approach can be explored. Such an automated approach would use automatic speech recognition (ASR) to "listen" to an individual producing spoken language and produce a phonetic transcription of the utterance, which would then be processed using the approaches described in this article. ASR technology is notoriously imperfect—especially in the face of speech-motor issues—but the field is improving rapidly, and there are examples of using even very imperfect ASR systems as part of language-assessment programs (Fraser et al., 2014; Roark et al., 2011; Sano et al., 2013). In its present form, without ASR, our system could certainly be used to automatically analyze existing data sets containing phonetic transcriptions, and could be used in clinical or research settings in which transcription is an option.

In the analyses presented in this article, the performance of each classifier was evaluated in isolation, on the basis of only the relevant productions and assuming perfect classification in the previous step. For example, the classification accuracy of the semantic-similarity criterion assumes that the phonological-similarity criterion was 100% accurate. However, even though the latter criterion was highly discriminant, some productions were in fact misclassified (e.g., some productions were erroneously tagged as phonologically related errors). If we had proceeded in a linear fashion and applied the semantic-similarity criterion to the results of the phonological-similarity analysis, the sample would have included some irrelevant items (e.g., phonologically unrelated errors set to be classified by the semantic-similarity criterion as either mixed or formal paraphasias). Chaining our classifiers together in this manner would cause these errors to accumulate, and so we would therefore expect the overall classification performance of the system to be somewhat lower in such a scenario. A more robust approach would be to use the results of our various classifiers as inputs to a larger system that would then combine all the available evidence and use machine-learning techniques to identify erroneous productions. Although the overall performance of the automated tools presented in this article is promising, much remains to be done before they will have fulfilled their potential.

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